

Reusable Crate Pilot at the Eastside Depot

Results and recommendation for the Operations Board, Tidewell Grocers. An invented sample: every name and number in it is made up.

Summary

This report describes the reusable crate pilot that ran at our Eastside depot this autumn. It presents the results of the pilot and discusses the options for our other three depots.

The pilot ran for 12 weeks, from September to November, and handled 18,400 orders. Drivers delivered groceries in plastic crates instead of cardboard boxes and collected the empty crates on their next visit. Customers paid a deposit of \$5 per crate, which we refunded when the crate came back.

Overall, customers responded well to the change. In the survey at the end of the pilot, 84% of the customers who answered said they preferred crates to boxes. Drivers needed some extra time at each stop, and a small share of crates was lost or damaged.

The last two sections set out the risks and costs of a full rollout and our recommendation to the board.

Background

Tidewell Grocers delivers around 1,500 grocery orders a day from four depots, and at the moment every one of those orders goes out in single-use cardboard boxes, which we buy from an outside supplier at a price that has risen steadily over the last three years and now makes up a noticeable share of the cost of each delivery.

Last year we spent \$1.9 million on boxes. The price per box rose by 22% over the same period, and the supplier has told us to expect a further rise next year. Boxes also cause problems at the door. They tear when they get wet, and customers have to break them down and put them out for recycling.

Packaging waste also matters to our customers. Most shoppers now say they would switch to a grocer that uses less packaging. It is very clear that this issue will only become more important going forward.

In addition, the company has set itself a target to halve its packaging waste by 2030. Boxes are by far the largest part of that waste.

Several of our competitors already deliver in reusable totes. The idea is simple. The totes go out full and come back empty with the next delivery, and after washing they are used again. A single tote can make hundreds of trips before it wears out.

In order to find out whether reusable crates would work for Tidewell, the operations team proposed a pilot at one depot. The board approved the pilot in June, with a budget of \$61,000.

The pilot had to answer three questions. Would crates cost less per order than boxes, once washing and losses were counted? Would customers give the crates back? And would the extra work at each stop be manageable for drivers?

How the pilot worked

In this section we will describe how the pilot worked: the crates, the deposit, how drivers collected empty crates, and how the depot washed them.

The pilot ran at the Eastside depot, which serves about 225 orders a day. We chose Eastside because it is our smallest depot and has spare floor space for a washing station. The pilot ran for twelve weeks, from the first week of September to the last week of November.

The depot bought 4,200 crates from a single supplier at \$9.50 each. Each crate holds about as much as two standard boxes and has a lid that folds shut. A typical order fills two crates. Customers paid a deposit of \$5 per crate.¹

The supply agreement covers breakage. Clause 7.2 reads: "The Supplier shall replace, at no cost to the Buyer, any crate that fails in normal use within twenty-four months of delivery."

Drivers collected empty crates when they made the customer's next delivery. Customers left the empty crates at the door, and the driver scanned each crate before loading it into the van. Crates that had not come back within 30 days were counted as lost, and the customer's deposit was kept.

At the depot, returned crates went through a washing station that we leased for the pilot. A team of 3 staff washed, dried and checked the crates each morning, and clean crates were back on the shelves by midday.

The crates held up well. By the sixth week, only 2% of the crates in use had been damaged, and none had failed in a way that the supplier would have to replace.

We told customers about the pilot by email a week before it started. The email explained the deposit and how to return crates, and it offered customers the choice of staying with boxes. Only 3% of customers chose to stay with boxes, and they are left out of the results that follow.

We measured the pilot in three ways. The depot's order system recorded every crate that went out and came back. The finance team worked out the cost per order each month. At the end of the pilot, we sent a short survey to every customer who had received crates.

Results

Cost per order

Over the 12 weeks, the depot delivered 18,900 orders in crates. The packaging cost per order fell from \$1.42 with boxes to \$0.98 with crates, a fall of 31%. The crate figure includes washing, the lease of the washing station, lost and damaged crates, and the price of the crates spread over their working life.

¹ The deposit was added to the order total and refunded to the customer's account when the crate was scanned back in at the depot.

Cost per order	Boxes	Crates
Boxes or crates	\$1.42	\$0.10
Washing and station lease	none	\$0.52
Lost and damaged crates	none	\$0.36
Total	\$1.42	\$0.98

The cost per order fell a little each month as the washing team got faster. It was \$1.05 in September, \$0.97 in October and \$0.93 in November. This result shows the wider promise of reuse for the whole business.

It is important to note that the price of the crates is basically spread over a working life of 200 uses, which is the figure the supplier gives, and if the crates wear out sooner than that in real use then the cost per order would be higher than the figure in the table.

Returns and losses

Customers returned 96% of crates within 30 days, and most came back with the next delivery. The other 4% were counted as lost. The deposits we kept covered a little over half of the cost of replacing them. On the first Monday, the washing station choked on a whole weekend of returned crates.

Drivers collected empty crates when they made the customer's next delivery. Customers left the empty crates at the door, and the driver scanned each crate before loading it into the van. Crates that had not come back within 30 days were counted as lost, and the customer's deposit was kept.

In addition, damage was lower than we expected. By the end of the pilot, 3% of crates had cracked sides or broken lids, and the supplier replaced all of them under the supply agreement.

Customer response

At the end of the pilot, 1,140 customers answered a short survey. Of these, 81% said they preferred the totes to boxes, 12% had no preference, and 7% preferred boxes. The reasons given most often were less waste and not having to break down boxes.

The most common complaint was about space. Some customers had nowhere to keep empty crates between deliveries, especially in small apartments. I think a crate that folds flat would answer most of these complaints.

Driver time

Collecting and scanning crates added about 40 seconds to each stop, which over a typical route of 28 stops adds up to almost 19 minutes a day, and drivers told us that they could leverage the extra time at the door to check that the customer had received everything they ordered.

Drivers were slower in the first two weeks, while customers got used to leaving crates out. After that, the extra time per stop settled at about 40 seconds. No routes ran late because of the crates.

In summary, as we have seen above, crates cost less per order than boxes, most customers gave them back, and the extra time at each stop was manageable. The next section looks at what customers said about the crates.

Risks and costs of a full rollout

A rollout to all four depots would need about 28,000 crates, three more washing stations and some changes to the shelving in our vans. It also brings risks that the pilot could not fully test.

The main risk is loss—crates that customers keep, throw away or use for something else. At Eastside, 4% of crates were lost. At our larger depots, where more orders go to apartment buildings, the loss rate could be higher.

We can reduce this risk in three ways: raise the deposit, remind customers by text message when a crate is overdue, and collect crates from customers who stop ordering.

The second risk is washing capacity. The leased station at Eastside ran at about 70% of its capacity. The larger depots would need bigger stations or a second washing shift. Washing also uses water: the Eastside station used about 1.2 liters per crate. Earlier studies found that washing uses less water than making new boxes.

The third risk is the supplier. The pilot used a single supplier, and the supply agreement runs for only two years.

Costs

The upfront cost of a full rollout is \$304,000: \$266,000 for crates, \$30,000 for three washing stations and \$8,000 for van shelving. At the pilot's cost per order, the rollout would save about \$0.44 per order, or about \$19,800 a month across the four depots. At that rate, the upfront cost would be paid back in about 15 months.

Options we rejected

We looked at two other options. The first was to switch to a cheaper cardboard box. This would cut packaging costs by about 8%, but it would do nothing about waste. The second was to use reusable bags instead of crates. Bags cost less to buy, but they are harder to wash and to stack in vans.

Recommendation and next steps

We recommend rolling out reusable crates to all four depots from April.

The pilot cut the packaging cost per order by 31%, customers returned 96% of crates, and most customers preferred crates to boxes. The rollout needs \$304,000 up front and would pay for itself in about 15 months.

We ask the board to approve this budget at its March meeting.

If the board agrees, the operations team will order the crates in February and install the washing stations in March. The rollout would start at the North depot in April, and the River and Hill depots would follow in May and June. Eastside would keep its crates from the pilot.

In addition, the team will test a crate that folds flat at Eastside from April, to answer the complaints about storage space. We will report the loss rate, the cost per order and customer feedback to the board each month for the first six months.